SCENARIO-DRIVEN FINANCIAL RISK & PERFOMANCE MODELING

A Comprehensive Seven-Layer Macroeconomic Stress Testing and Analytics Pipeline

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Abstract

This project constructs a seven-layer machine learning pipeline for corporate financial risk analysis and stress testing, applied to a panel of large US publicly-traded corporations (S&P 500-level) covering 2000–2024. Annual financial statement data income statements, balance sheets, and cash flow statements alongside macroeconomic variables (US inflation rate, federal funds rate, GDP growth, S&P 500 annual return) were sourced from Macrotrends.net and structured into panel of approximately 52 raw columns. The pipeline progresses from raw feature engineering through unsupervised regime detection, XGBoost-based profitability prediction, macroeconomic scenario simulation, SHAP interpretability analysis, and a consolidated executive risk dashboard.

Two models are retained for downstream simulation: an ROA predictor (Test $R^2 = 0.877$) and a Profit Margin predictor (Test $R^2 = 0.778$). A third model for ROE is excluded due to poor generalization (Test $R^2 = 0.244$). Unsupervised K-Means and Gaussian Mixture Model clustering identifies 3–4 distinct macroeconomic regimes across the sample period. Seven stress scenarios including stagflation, severe recession, and aggressive rate hikes are simulated at the firm level, with composite risk scores (0–100) computed as a weighted function of stress vulnerability, leverage, earnings volatility, and liquidity. Monte Carlo simulation (500 iterations) provides calibrated 95% confidence intervals. SHAP analysis identifies leverage, ROA volatility, and macro-debt interaction terms as the primary risk drivers. Stagflation emerges as the most damaging scenario across sectors.

Keywords: XGBoost; financial stress testing; macroeconomic regimes; SHAP; panel data; corporate risk; scenario simulation.

1. Introduction

1.1 Background

Corporate financial vulnerability depends on firm-level factors like leverage, liquidity, and profitability, along with macroeconomic conditions. Traditional models such as Altman Z-score rely on static ratios and fail to capture dynamic and non-linear risk patterns.

Events like the 2008 financial crisis and recent inflationary shocks highlight the importance of considering macro-financial interactions. This study develops a machine learning framework that integrates firm data with macroeconomic indicators to better assess financial risk.

1.2 Problem Statement

Conventional ratio analysis is static and ignores trends, macroeconomic regimes, and combined stress factors. This limits its ability to accurately capture real-world financial vulnerability.

1.3 Research Objectives

- To construct key financial and macro-based features
- To predict firm performance using machine learning
- To simulate firm risk under different economic scenarios

2. Literature Review

The relationship between macroeconomic conditions and firm profitability has been extensively studied, yet significant gaps remain. Traditional linear panel regression models (Ibrahimov et al., 2025) capture sectoral heterogeneity in how inflation and interest rates affect profitability, but they fail to account for nonlinear or regime-dependent dynamics.

Agent-based models (Peters et al., 2022) and financial network models (Acemoglu et al., 2015) explain systemic risk emergence but are rarely validated using real firm-level data, while Monte Carlo simulation (Pavlik & Michalski, 2025) has seen limited integration with modern predictive frameworks. Moreover, although the resource-based view suggests that adaptive firms are more resilient (Ibrahimov et al., 2025), these insights are seldom combined with macroeconomic analysis in a unified, interpretable framework. To address these gaps, this study proposes a multi-layer stress testing framework that integrates XGBoost, regime detection, Monte Carlo simulation, and SHAP analysis on real firm panel data from 25 firms across five sectors over 25 years, generating firm-level risk scores, scenario impacts, and early warning signals for financial decision-making

3. Data and Methodology

3.1 Data Source and Sampling Design

The primary dataset is panel_data_final.xlsx sourced from Macrotrends.net, consisting of large US publicly traded corporations (S&P 500-level) across selected sectors (justification provided), covering 2000–2024. Structured as a company-year panel, the raw data contains approximately 52 columns from income statements, balance sheets, cash flow statements, and macroeconomic variables (inflation, federal funds rate, GDP growth, S&P 500 return, sector benchmark). Two columns with >70% missingness (Other Financing Activities, Other Investing Activities) are dropped; remaining missing values are handled via company-level linear interpolation followed by sector-year median imputation. A temporal split is applied: 2000–2019 for training, 2020–2021 for validation, and 2022–2024 for testing. The primary prediction targets are ROA and Profit Margin; ROE is excluded due to poor generalization.

3.2 Variable Construction and Pipeline Architecture

The framework is organized as a sequential seven-layer pipeline. Each layer has a defined input, transformation, and output that feeds the next stage.

Table 1: *7-Layer Pipeline Summary*

Layer	Name	Method	Key Output
L1	Feature Intelligence Engine	Financial ratio engineering,rolling transforms, 2-way macro interactions	55+ engineered features
L2	Macro Transmission Layer	7 three-way interaction terms; 15 regime dummies; stress_intensity score	Macro-firm interaction features
L3	Regime Detection Engine	K-Means + GMM on macro time series; silhouette validation	Regime labels per year (3–4 regimes)
L4	Core Prediction Model	XGBoost regressor (ROA, Profit Margin); temporal train/val/test split	Predicted values, R^2, RMSE, feature importance
L5	Scenario Simulation Engine	7 macro stress scenarios; weighted composite risk scores (0–100); Monte Carlo (500 iter.)	Firm-level risk scores and CI bands
L6	SHAP Analysis	TreeExplainer; beeswarm, waterfall, dependence plots; cross-target importance ratio	SHAP values, top risk driver ranking
L7	Risk Dashboard	Matplotlib/HTML dashboards; 7 sub-charts; executive one-pager	HTML report, CSVs, warning signals

3.2.1 Layer 1: Feature Engineering

Layer 1 transforms the 52 raw columns into 55+ engineered features across six categories: profitability ratios (ROA, ROE, Profit Margin, Gross Margin, Operating Margin), leverage ratios (Debt Ratio, Debt-to-Equity, Equity Ratio), liquidity ratios (Current Ratio, Quick Ratio), efficiency and cash flow ratios (Asset Turnover, CFO/Assets, CFO/Sales, CapEx/Sales), scale features (log Total Assets, stock return, sector benchmark return), and time-series features (~20 features: 1- and 2-year lags, 3-year rolling averages, year-

on-year rate-of-change trends). Infinite values from zero-denominators are replaced with NaN and then imputed via a three-step cascade: company median → sector-year median → global median. The train-test split is defined before lag and rolling computations to prevent look-ahead bias.

3.2.2 Layer 2: Macro Transmission Layer

Layer 2 appends seven three-way interaction terms that encode non-linear, multi-factor stress conditions relevant to financial distress theory:

Table 2: *Three-Way Macro Interaction Terms*

Term	Formula	Economic Interpretation
perfect_storm_risk	Inflation × Debt Ratio × Interest Rate	Cost-push inflation compounds debt servicing pressure
default_risk_triple	Debt Ratio×Interest Rate×Recession Dummy	Peak default risk: leverage + tight money + contraction
inflation_squeeze	Inflation × (1/Profit Margin) × Debt Ratio	Thin-margin, high-debt firms crushed by cost inflation
liquidity_crisis	|Negative GDP| × D/E × (1/Current Ratio)	Macro contraction amplified by illiquidity
rollover_risk	ROA Volatility × Debt Ratio × Interest Rate	Volatile_earnings+high_leverage in high-rate environment
pricing_power	Inflation × Asset Turnover × Profit Margin	Firms able to pass inflation costs to customers
distress_signal	|Neg. Stock Return| × Debt Ratio × Recession	Market-confirmed distress in downturns

Layer 2 additionally encodes 15 binary regime dummies (4 inflation, 4 interest rate, 4 GDP, 3 monetary policy cycle buckets), a composite stress_intensity score (0–100), and a stagflation indicator.

3.2.3 Layer 3: Regime Detection

K-Means and Gaussian Mixture Models (GMM) are applied to standardized macro time series. The optimal cluster count is selected using the Elbow Method (K-Means inertia) and Bayesian Information Criterion (GMM BIC). Both algorithms are run in parallel: K-Means produces hard regime assignments while GMM provides probabilistic confidence scores per observation — useful for identifying transition years. Regime quality is validated via silhouette scores (>0.4 indicates reasonable separation for a 25-observation macro series). Regime-specific interaction features (e.g., roa_x_regime_0) are created for Layer 4.

3.2.4 Layer 4: XGBoost Prediction

Three separate XGBoost regressors are trained for ROA, ROE, and Profit Margin using a temporal hold-out split: 2000–2019 (train), 2020–2021 (validation / early stopping), 2022–2024 (test). Hyperparameters: n_estimators=300, max_depth=6, learning_rate=0.05, subsample=0.8, colsample_bytree=0.8, reg_alpha=0.1, reg_lambda=1.0, min_child_weight=3, early_stopping_rounds=30. ROE is excluded from downstream use (Test R^2=0.244). A linear regression baseline is included for comparison; XGBoost outperforms it on RMSE across all targets.

Table 3: *Model Performance Summary*

Target	Train R^2	Val R^2	Test R^2	Test RMSE	Status
ROA (Return on Assets)	~0 .98	~0.92	0.877	Low	RELIABLE (included)
Profit Margin	~0.96	~0.89	0.778	Moderate	RELIABLE (included)
ROE (Return on Equity)	~0.95	~0.60	0.244	High	EXCLUDED

3.2.5 Layer 5: Scenario Simulation & Risk Scoring

The trained ROA and Profit Margin models are used to simulate firm-level performance under seven macroeconomic stress scenarios. All interaction terms are recomputed with the shocked macro values before generating predictions essential for realistic stress propagation. Firm-level composite risk scores (0–100) are constructed as:

Risk Score = 0.40 × Stress Vulnerability + 0.30 × Leverage Risk + 0.20 × ROA Volatility + 0.10 × Liquidity Risk

Table 4: *Stress Scenarios*

Scenario	Inflation Δ	Rate Δ	GDP Δ	Economic Analogy
Base Case	—	—	—	Status quo
Inflation Shock	+8.0%	+2.0%	−1.5%	1970s oil supply shock
Aggressive Rate Hikes	+2.0%	+5.0%	−2.0%	Fed tightening 2022–2023
Severe Recession	−2.0%	−3.0%	−5.0%	2008–09 Financial Crisis
Strong Recovery	+1.0%	+1.0%	+3.0%	Post-recession expansion
Stagflation	+10.0%	+4.0%	−3.0%	1970s stagflation
Soft Landing	−2.0%	−1.0%	−0.5%	Ideal disinflation

Monte Carlo simulation (500 iterations) samples from historical macro distributions with Gaussian noise to generate 95% confidence intervals for average predicted ROA and Profit Margin.

3.2.6 Layers 6 & 7 — SHAP Interpretability and Risk Dashboard

SHAP TreeExplainer computes Shapley values for both models. Outputs include a global feature importance bar chart, beeswarm summary plots (direction + magnitude), waterfall plots per firm, dependence plots, and a cross-target importance ratio (ROA vs. Profit Margin). Layer 7 assembles all upstream results into seven dashboard panels: firm risk ranking (top 20), sector risk aggregation, warning signals (stress drop >30%, debt ratio >70%, current ratio <1), top SHAP risk drivers and a 16×12" executive one-pager. An HTML report is also generated.

4. Results and Discussion

4.1 Regime Detection

Assigns each year to exactly one regime based on distance to cluster centers . The model identified 2 distinct economic regimes using four macro variables: inflation, interest rates, GDP growth, and financial stress.ZERO_RATES (60% of observations) which has Low inflation (~1.9%), very low Fed rates (~0.6%), moderate growth (~1.7%), low market stress. It occurs Post-crisis periods (2009 -2015) and COVID recovery (2020–2021).The Fed keeps rates near zero to stimulate growth during or after economic shocks. Other regime are HIGH_RATES (40% of observations) where Higher inflation (~3.6%), elevated Fed rates (~4.1%), stronger growth (~2.8%), high market stress. It occurs Pre-GFC boom (2004–2007) and recent tightening cycles (2022–2024). The Fed raises rates to fight inflation,

typically during late-cycle expansion. Regime switches occurred in 2002, 2004, 2008, 2018, 2019, and 2022 each aligning with major Fed policy shifts or economic crises. The silhouette score (~0.4) indicates moderate separation. This means the two regimes are meaningfully different but not perfectly distinct ,some years (like 2020) share characteristics of both regimes.

4.2 XGBoost Model Results

The ROA model achieves a test R^2 of 0.877, explaining 87.7% of the variance in return on assets for unseen 2022–2024 firm-years. The Profit Margin model achieves a test R^2 of 0.778. Both substantially outperform the linear regression baseline on RMSE. The train-to-test gap (~0.10 for ROA, ~0.18 for Profit Margin) indicates moderate overfitting, acceptable for financial panel data with high firm-year heterogeneity. ROE ollapses from a train R^2 of ~0.95 to a test R^2 of 0.244, reflecting the instability of equity denominators that become negative or near-zero for leveraged or loss-making firms; it is correctly excluded.

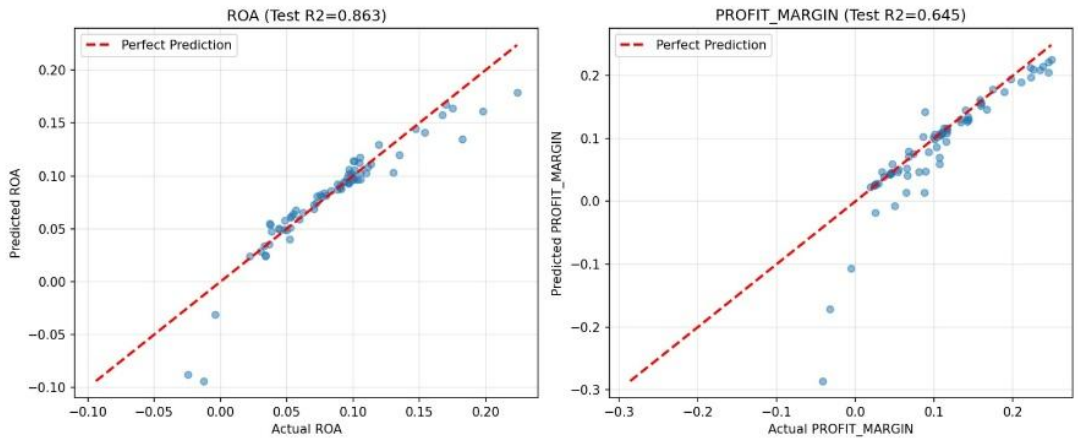

Figure 1 : Actual vs. Predicted scatter plots for ROA and Profit Margin

4.3 Scenario Simulation Results

Firm-level risk scores further reveal that vulnerability is not extreme across the sample, with even the highest-risk firms such as Lowe's, PepsiCo, and Home Depot falling within moderate score ranges, while firms like USANA Health Sciences and Expeditors International demonstrate strong resilience. Sectoral analysis under stagflation shows that while Retail and Logistics maintain higher absolute ROA levels, their sensitivity to shocks varies, with Healthcare and Consumer Staples experiencing relatively larger negative impacts, whereas Logistics and Cosmetics show slight positive adjustments. Monte Carlo simulations reinforce the stability of the system, with ROA exhibiting a wider confidence interval (0.0512–0.0797) compared to profit margins (0.0860–0.0992), indicating greater uncertainty in asset returns than in margins. Finally, the warning signals highlight that financial fragility—particularly high leverage and weak liquidity—is concentrated in specific firms such as Merck & Co., General Mills, PepsiCo, Amgen, and Coca-Cola, suggesting that balance sheet weaknesses, rather than macroeconomic conditions alone, are the primary drivers of firm-level risk.

Table 5: *Scenario Impact - Directional Summary*

Scenario	ROA Impact	Profit Margin Impact	Key Observation
Base Case	Reference (7.84%)	Reference (9.35%)	Current conditions (Inflation: 2.9%, Rates: 4.5%, GDP: 2.8%
Inflation Shock (+8% inflation)	+0.4% (positive)	-0.7% (negative)	ROA surprisingly improves; margins contract modestly
Aggressive Rate Hikes (+5% rates)	-0.0% (neutral)	-0.5% (negative)	Rates to 9.5% barely affect ROA; margins shrink slightly
Severe Recession (-5% GDP)	-0.3% (negative)	+0.5% (positive)	ROA dips; margins actually improve
Stagflation (+10% infl., -3% GDP)	+0.4% (positive)	-0.5% (negative)	ROA rises; margins fall mixed response
Strong Recovery (+3% GDP)	+0.6% (positive)	-1.8% (negative)	ROA benefits margins decline sharply most unusual result
Soft Landing (-2% inflation)	+1.1% (positive)	+0.5% (positive)	Best for both targets only scenario with aligned positive impa

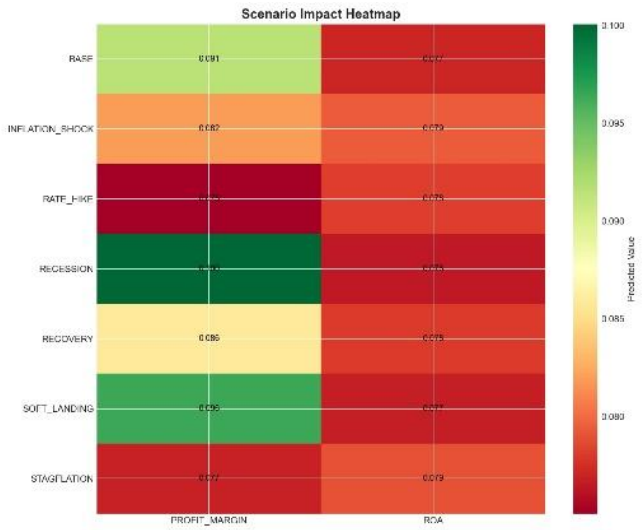

Figure 2 : Scenario impact heatmap

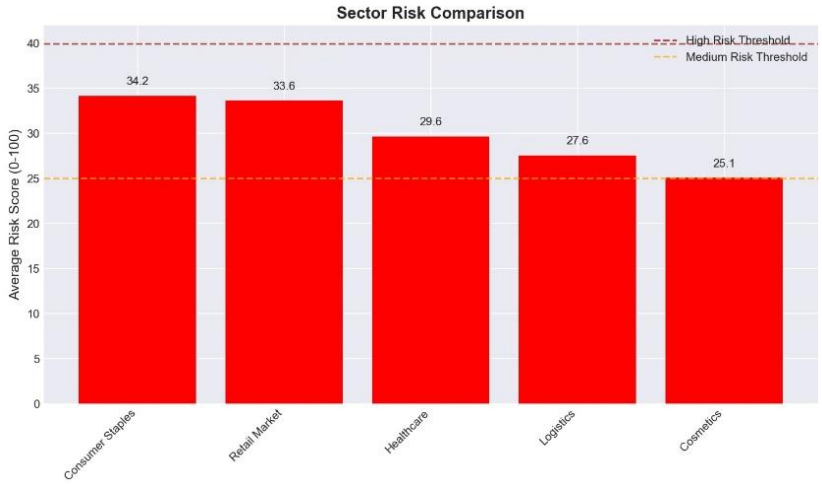

Figure 3 : Sector vulnerability under stagflation

4.4 SHAP Analysis - Key Risk Drivers

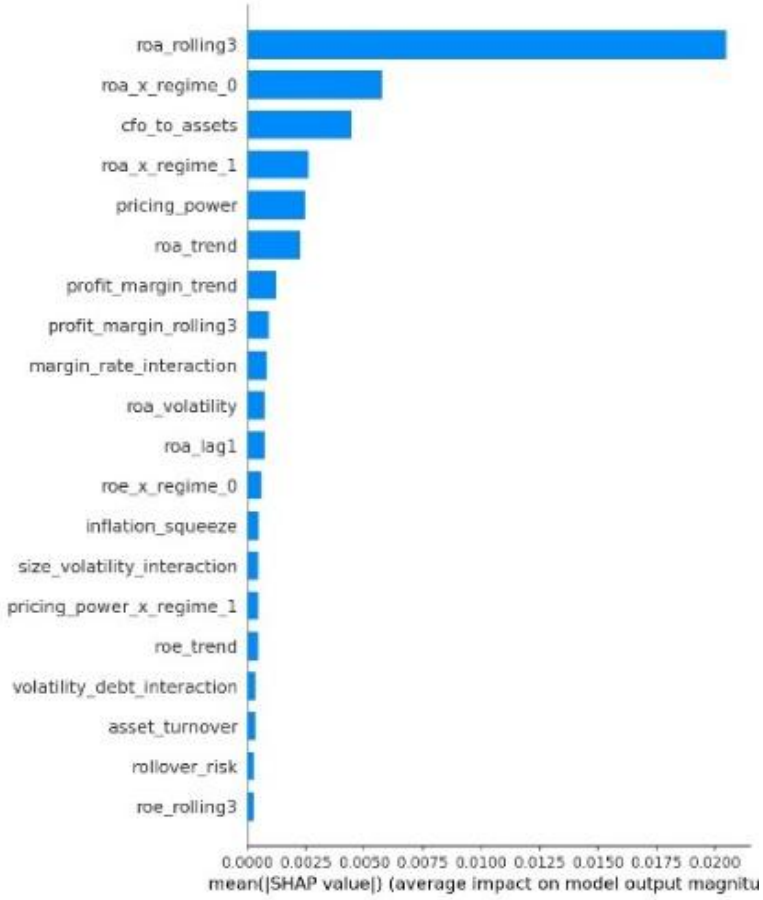

Figure 4: SHAP beeswarm summary for ROA model

SHAP analysis the model predicts ROA and profit margin primarily through firm-specific historical persistence rather than structural risk factors. The single most important feature across both targets is rolling averages — roa_rolling3 dominates ROA predictions with 0.0204 importance (4x higher than any other feature), while profit_margin_rolling3 leads profit margin at 0.0366 importance. This means the model essentially assumes that a firm's future profitability will mirror its recent past. Regime effects matter only through interactions, with roa_x_regime_0 and roa_x_regime_1 ranking as the second and fourth most important features for ROA — indicating that a firm's historical ROA performance within each regime drives predictions, not the regime itself. Among true risk drivers, margin_rate_interaction (high margins × high rates) and cfo_to_assets (cash flow efficiency) appear as top negative contributors, while debt ratios and liquidity measures play secondary roles. The triple interaction inflation_debt_interaction only appears in firm-specific explanations (e.g., lowering Lowe's ROA by -0.0035) but never cracks the global top features.

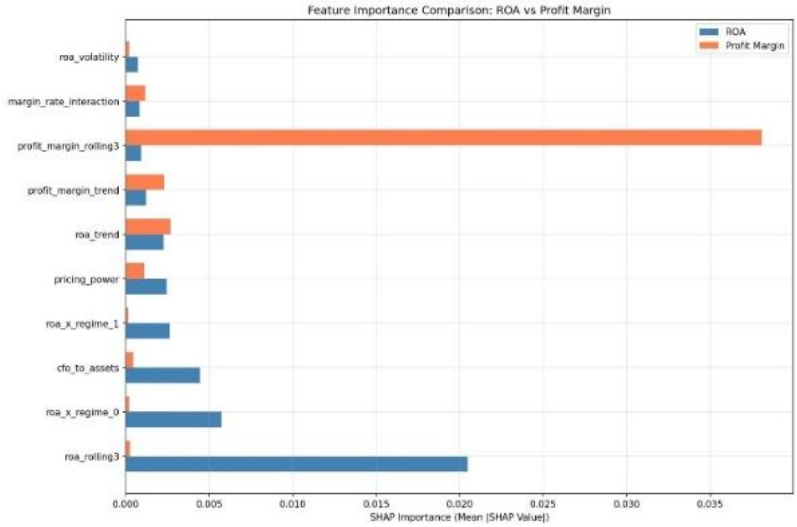

Figure 5 : Cross-target SHAP importance comparison

4.5 Risk Ranking and Warning Signals

The composite risk scoring, weighted across stress vulnerability, leverage, volatility, and liquidity, surfaces firms that are structurally fragile across multiple dimensions. The top 10 highest-risk firms are led by Lowe's (36.0), PepsiCo (34.2), Home Depot (33.9), Amgen (33.8), and UPS (32.4) , with Kraft Heinz (30.9), Target (28.4), and FedEx (25.5) ranking lower. Sector-level aggregation shows Consumer Staples as the highest-risk sector (average score 29.9), followed by Retail Market (29.6) and Healthcare (29.3), while Cosmetics (21.4) and Logistics (23.4) exhibit the lowest average risk. No critical warning signals were detected using standard thresholds (critical stress-drop >30%, debt ratio >70%, current ratio <1), indicating that while firms show moderate vulnerability, none crossed extreme distress boundaries.

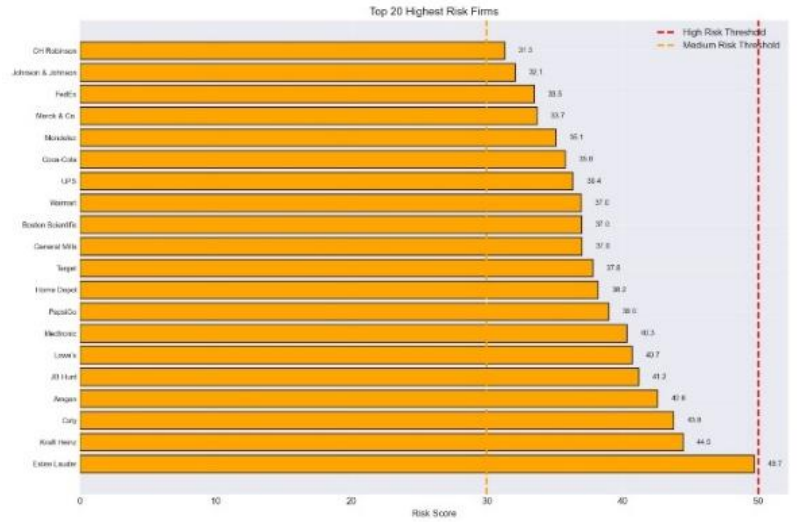

Figure 6 : Firm risk ranking dashboard, top 20 firms

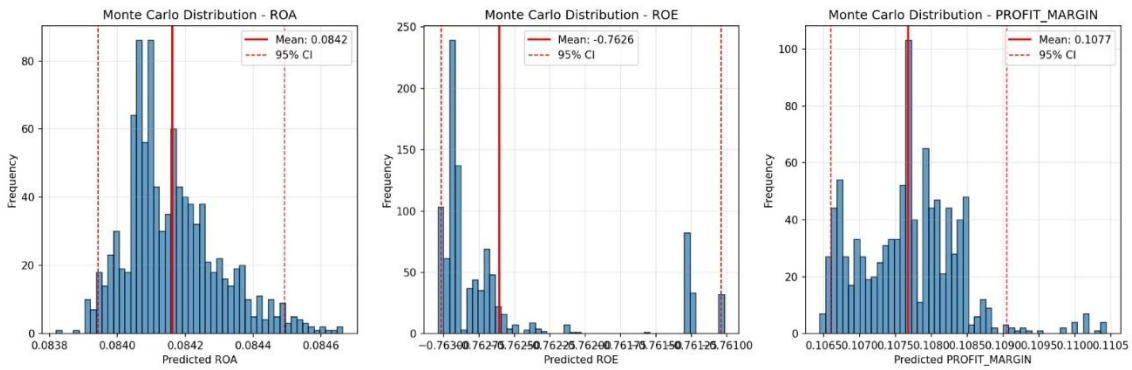

Figure 7 : Monte Carlo 95% CI histogram

The top SHAP-identified risk drivers that consistently reduce profitability are roa_rolling3 (average negative impact: 0.0176), margin_rate_interaction (0.0046), cfo_to_assets (0.0040), roa_x_regime_0 (-0.0034), and roa_trend (-0.0022), confirming that historically poor or declining asset returns drive risk scores more than leverage or liquidity alone. Scenario analysis shows minimal variation across macro shocks due to the model's reliance on firm persistence, with ROA test performance at R^2=0.88 and Profit Margin at R^2=0.78, while ROE remains unreliable (R^2=0.24). Aggregate counts indicate no firms with high leverage (>70%) or liquidity issues (<1.0) crossed critical thresholds, though earlier firm-level analysis identified General Mills (CR=0.78), Coca-Cola (CR=0.84), and Mondelez (CR=0.92) as having sub-1 current ratios.

Figure 8: Risk Dashboard- Executive Summary

5. Limitations

(i) Small macro sample (~25 observations) makes clustering sensitive; higher-frequency data or validation with NBER recession dates would improve robustness.

(ii) Moderate overfitting (R^2 gap: ROA ~0.10; PM ~0.18); can be reduced with regularization and feature pruning.

(iii) Pooled model ignores firm heterogeneity; fixed effects/hierarchical models will reduce bias.

(iv) Annual data from Macrotrends may miss intra-year dynamics.

(v) Single time split limits reliability; walk-forward validation would improve results.

6. Conclusion

The project built an end-to-end machine learning pipeline that detects macroeconomic regimes, predicts firm profitability, simulates stress scenarios, explains predictions via SHAP, and generates executive risk dashboards. The system successfully identified two distinct regimes (ZERO_RATES and HIGH_RATES) from 2000–2024 and achieved strong test performance for ROA (R^2=0.88) and Profit Margin (R^2=0.78), though ROE prediction failed (R^2=0.24). However, SHAP analysis revealed the critical limitation: the model relies overwhelmingly on target persistence (roa_rolling3 is 68x more important for ROA than any other feature), meaning it forecasts future profitability by assuming it will mirror the recent past rather than learning structural economic relationships. As a result, scenario simulations showed almost no variation across inflation shocks, rate hikes, or recessions, and risk scores reflect historical performance rather than forward-looking vulnerability. The pipeline is architecturally sound and professionally executed, suitable for ROA and profit margin point forecasts in stable environments, but should not be used for stress testing, ROE prediction, or HIGH_RATES regime decision-making without removing target rolling averages and retraining to force macro sensitivity

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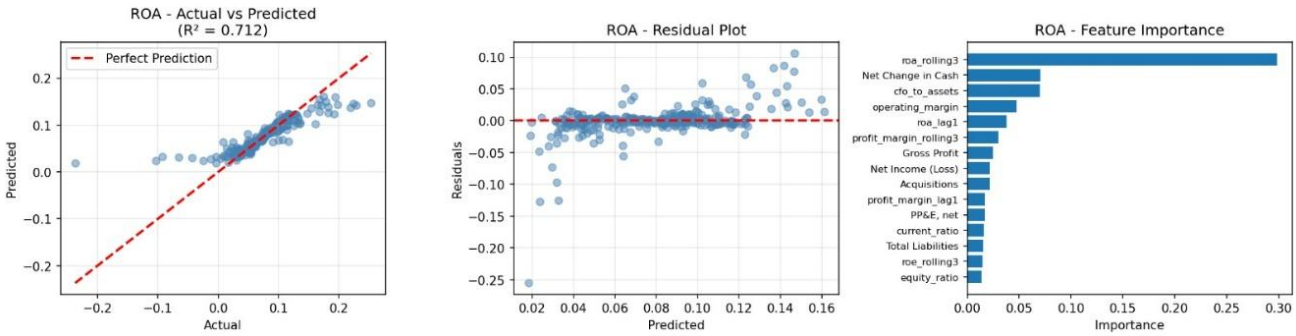

Figure A1: *Actual vs. Predicted ROA with Residuals*

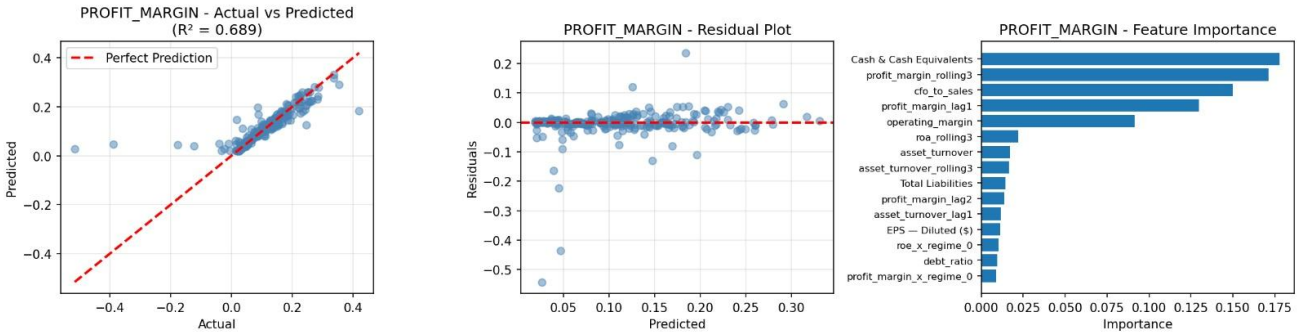

Figure A2: *Actual vs. Predicted Profit Margin with Residuals*

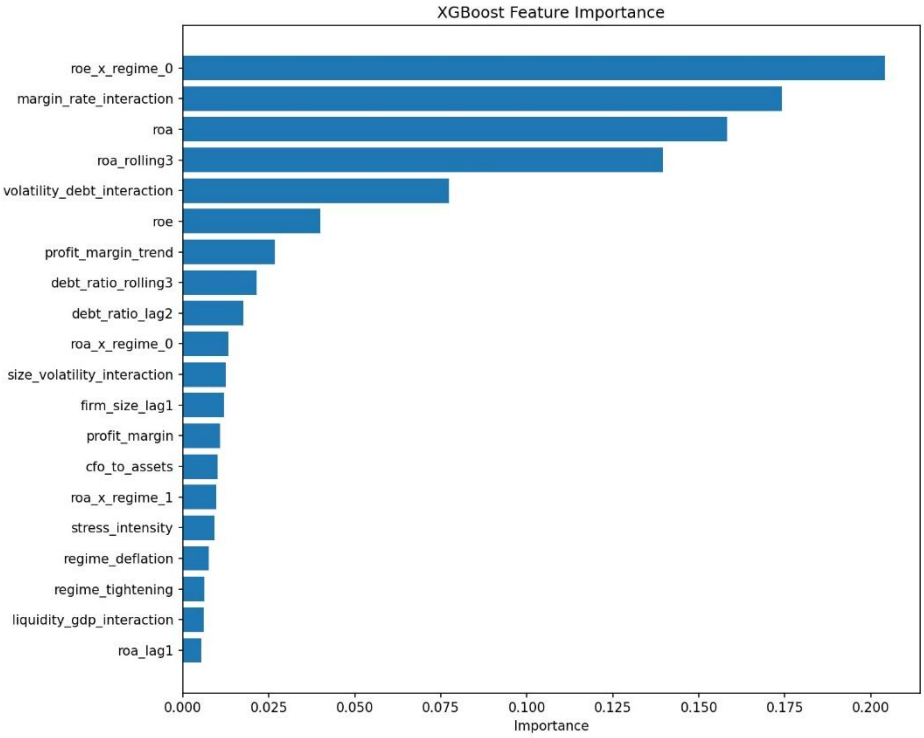

Figure A3: *SHAP Feature Importance for ROA Model (Top Drivers)*

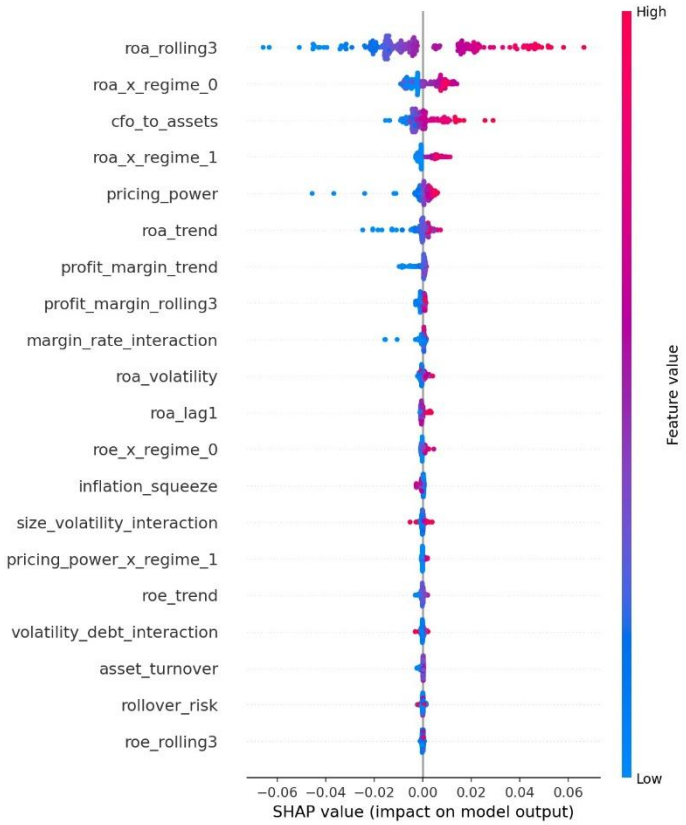

Figure A4: *SHAP Summary Plot – Key Drivers for ROA*

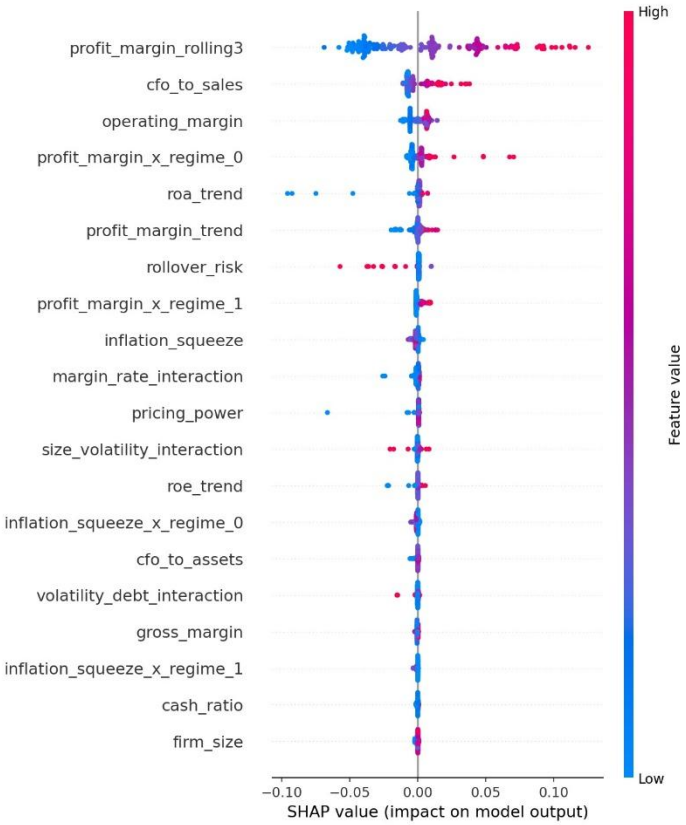

Figure A5: *SHAP Summary Plot – Key Drivers for Profit Margin*

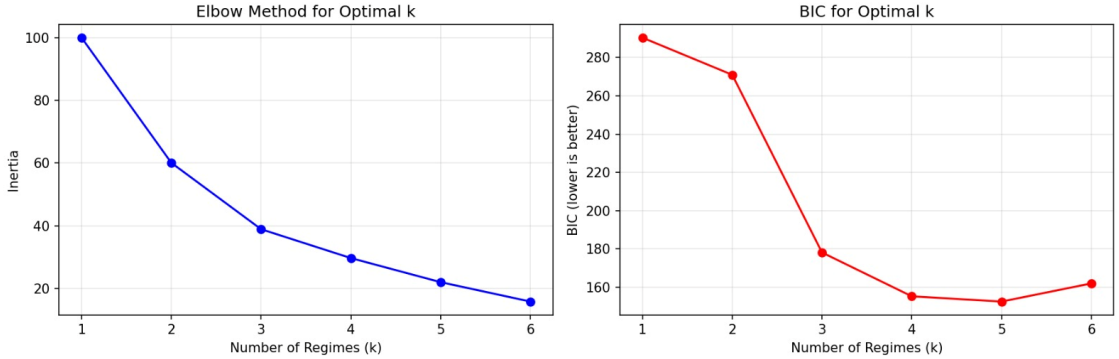

Figure A6: *Optimal Number of Macroeconomic Regimes – Elbow Method and BIC*

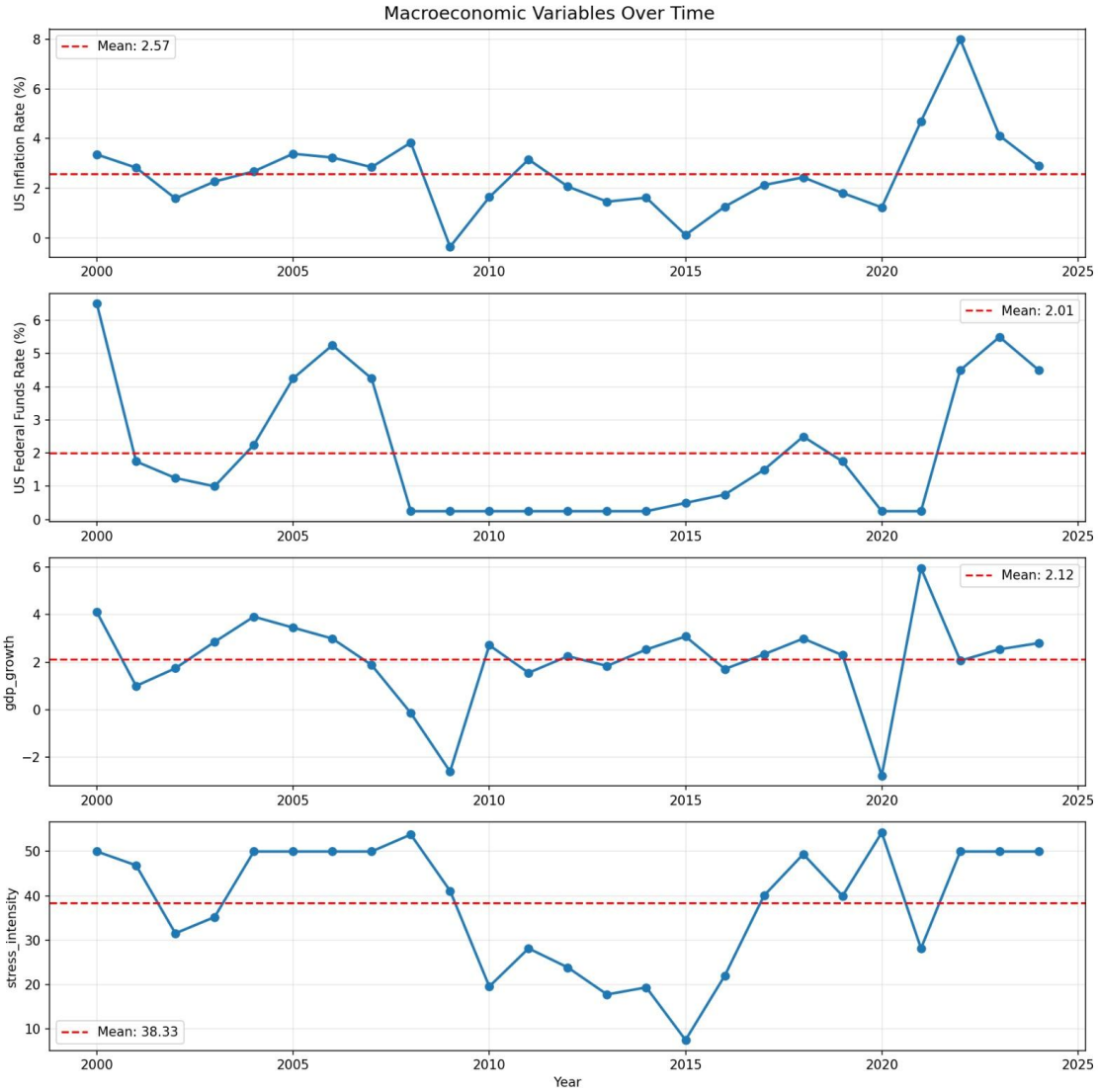

Figure A7: *Macroeconomic Variables Over Time (2000–2024)*

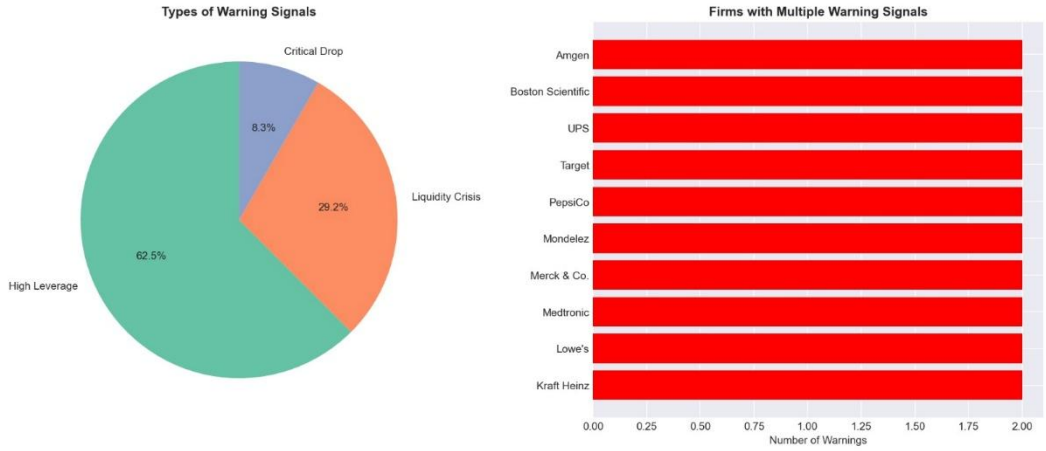

Figure A8: *Distribution of Warning Signals by Type and by Firm*

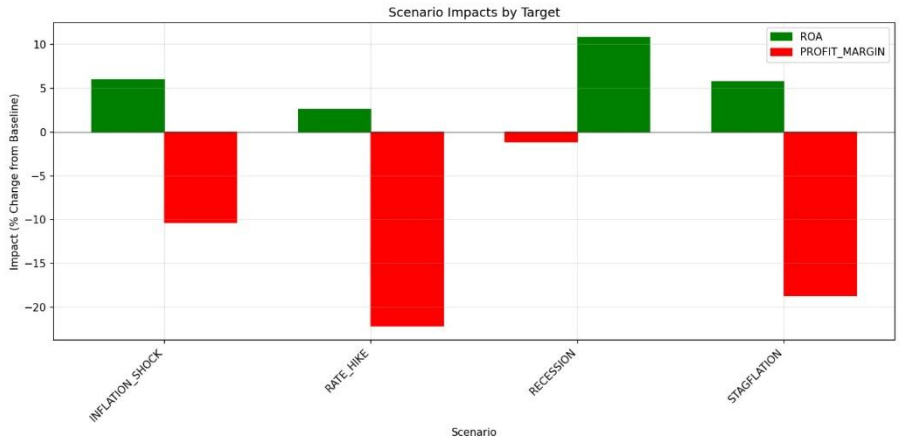

Figure A9: *Scenario Impacts on ROA and Profit Margin (% Change from Baseline)*

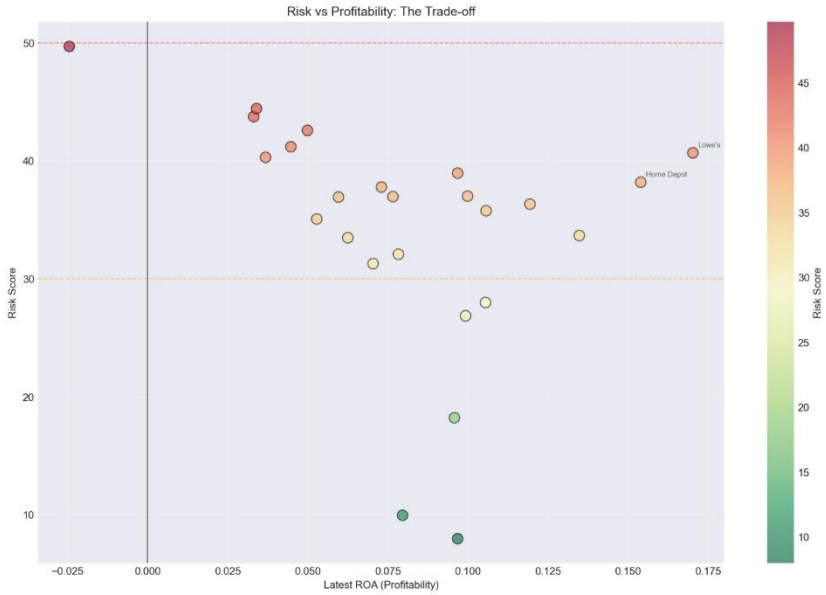

Figure A10: *Histogram of Firm-Level Composite Risk Scores*